

PROJECT
MOVIE TICKET BOOKING SYSTEM

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DATABASE SYSTEM
5(B)

DESCRIPTION:

The **Movie Ticket Booking System** is a database-driven application designed to manage movie screenings, customer bookings, theater information, and ticket reservations efficiently. The system allows users to browse movies, select showtimes, reserve seats, and make payments online.

The project demonstrates core database concepts such as:

- Entity Relationship Modeling (ERD)
- Normalization
- Primary and Foreign Keys
- Relationships (1:1, 1:M, M:N)
- SQL Queries
- Transaction Management

Main Objectives

- Store movie and theater information
- Manage show schedules
- Handle customer registrations
- Process ticket bookings
- Track seat availability
- Maintain payment records

Entities and Attributes

1. Customer

Stores customer details.

Attribute	Description
CustomerID (PK)	Unique customer ID
Name	Customer name
Email	Email address
Phone	Contact number

Attribute	Description
Password	Login password

2. Movie

Attribute	Description
MovieID (PK)	Unique movie ID
Title	Movie title
Genre	Action, Comedy, etc.
Duration	Movie length
Language	Movie language
ReleaseDate	Release date

3. Theater

Stores theater/cinema information.

Attribute	Description
TheaterID (PK)	Unique theater ID
TheaterName	Cinema name
Location	Theater address

4. Screen

Represents screens inside theaters.

Attribute	Description
ScreenID (PK)	Unique screen ID
TheaterID (FK)	Related theater
ScreenName	Screen number/name
TotalSeats	Seating capacity

Relationship

- One Theater can have many Screens
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5. Show

Stores movie show schedules.

Attribute	Description
ShowID (PK)	Unique show ID
MovieID (FK)	Related movie
ScreenID (FK)	Related screen
ShowDate	Date of show
ShowTime	Time of show
TicketPrice	Price per seat

Relationship

- One Movie can have many Shows
- One Screen can host many Shows

6. Seat

Stores seat information.

Attribute	Description
SeatID (PK)	Unique seat ID
ScreenID (FK)	Related screen
SeatNumber	Seat label
SeatType	Regular/VIP

Relationship

- One Screen contains many Seats

7. Booking

Stores ticket booking information.

Attribute	Description
BookingID (PK)	Unique booking ID
CustomerID (FK)	Related customer
ShowID (FK)	Related show

Attribute	Description
BookingDate	Date of booking
TotalAmount	Total payment
BookingStatus	Confirmed/Canceled

Relationship

- One Customer can make many Bookings
- One Show can have many Bookings

8. BookingSeat

Handles many-to-many relationship between Booking and Seat.

Attribute	Description
BookingSeatID (PK)	Unique ID
BookingID (FK)	Related booking
SeatID (FK)	Reserved seat

Relationship

- One Booking can reserve multiple Seats
- One Seat can belong to many bookings for different shows

9. Payment

Stores payment details.

Attribute	Description
PaymentID (PK)	Unique payment ID
BookingID (FK)	Related booking
PaymentMethod	Card/Cash/Online
PaymentDate	Payment timestamp
Amount	Paid amount
PaymentStatus	Successful/Pending

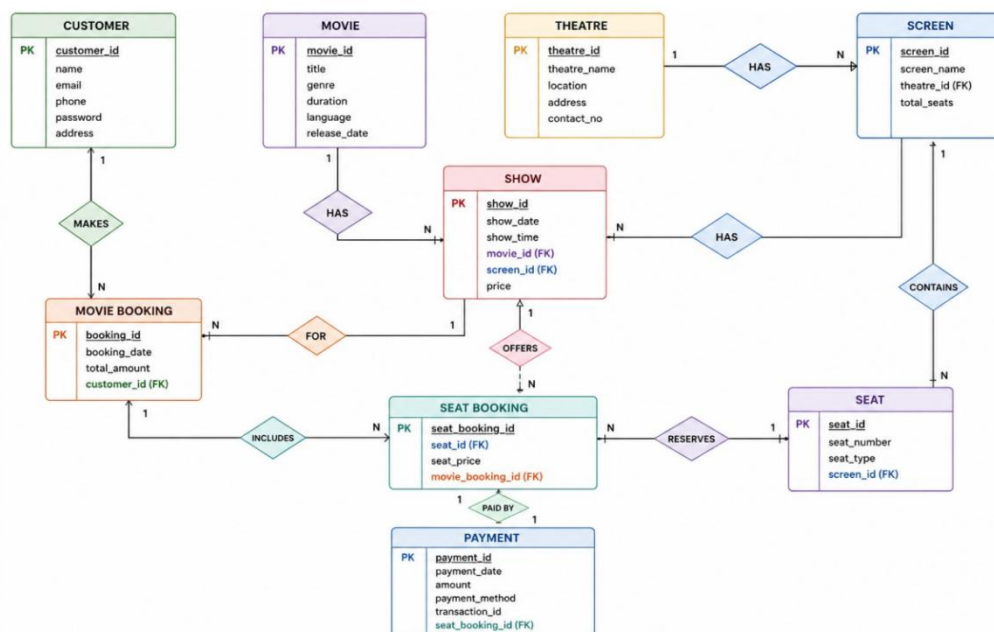
Relationship

- One Booking has one Payment

ERD Relationships Summary

Customer 1 ---- M Booking
Movie 1 ---- M Show
Theater 1 ---- M Screen
Screen 1 ---- M Seat
Screen 1 ---- M Show
Show 1 ---- M Booking
Booking 1 ---- 1 Payment
Booking M ---- M Seat (via BookingSeat)

ERD DIAGRAM:



EER DIAGRAM:

Enhanced Entity Relationship Diagram (EERD)
(Movie Booking System)

